



A thermal LBM model for studying complex flow and heat transfer problems in body-fitted coordinates



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ABSTRACT

A thermal lattice Boltzmann model in a body-fitted coordinate system has been proposed by combining the thermal lattice Boltzmann model on uniform grids with the generalized form of interpolation-supplemented lattice Boltzmann method (GILBM), and then it is validated by numerically simulating flow and heat transfer problems in domains with irregular geometry. The flow around a circular cylinder has been investigated under different Reynolds numbers. Both pressure coefficients and Nusselt numbers around cylindrical surface are in good agreement with previous numerical and experimental results. The natural convection in horizontal concentric and eccentric cylindrical annulus has also been studied, respectively. It is found out that equivalent heat conductivity increases gradually as Rayleigh number. What's more, the equivalent heat conductivities on the inner and outer cylindrical surfaces are consistent with the experimental results with acceptable deviations. These comparisons show that the proposed thermal lattice Boltzmann model is feasible and reliable to simulate fluid flow and heat transfer problems in domains with irregular geometry.

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1. Introduction

In recent years, the lattice Boltzmann method (LBM) as a novel method of computational fluid dynamics (CFD) has attracted much attention. It has been proved to be an effective and promising tool in the simulation of incompressible flow, porous media flow, and multiphase flow [1,2]. Together with modern computer of massively parallel processors, the advantages of LBM, which are the parallelism of the method, the simplicity of the programming and the capability of incorporating model interactions, have become more obvious. Although the LBM has such advantages and the particle distribution evolution equation can recover the Navier–Stokes equations through the Chapman–Enskog expansion [3,4], it has not been widely used as a practical tool of CFD due to the limitation of uniform grids in the standard LBM model. For most of practical problems, the computational domains usually have

irregular geometry. However, the grids of standard LBM model must be symmetrical and uniform which are impossible to be exactly located at the irregular boundaries. As a result, it is difficult to accurately implement the boundary conditions unless sufficient lattice nodes are used [5,6].

Such a restriction is directly inherited from the Lattice Gas Automaton (LGA), the ancestor of the LBM model. In the LGA, particle is of Boolean type and particles collision must take place at grid sites, so a regular grid is extremely needed to ensure that the particle can arrive to the adjacent nodes exactly at next time step, however, the Boolean particle population in the LGA has been replaced by a real-number single-particle distribution in the LBM. With such a real-number representation, the function of particle distribution can be regarded as continuous in the physical domain and its value can also be determined wherever it is in the computational domain [7,8]. Given this characteristic of LBM model, the limitation of uniform grids in the standard LBM model turns to be redundant and LBM model requires some improvements so as to be more suitable for practical applications.

To some degree, increasing the grids density can compensate for this defect, but it will unavoidably reduce the computational efficiency. Alternatively, in order to improve the computational efficiency, adaptive mesh refinement technique is incorporated

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Nomenclature

a	interpolation coefficient
c_s	lattice sound speed
D	diameter
dr	distance
D2Q9	2-dimension 9-velocity lattice
$\mathbf{e}$	lattice velocity
$\tilde{\mathbf{e}}_i$	contravariant velocity
f	particle distribution function
F	body force
$\mathbf{G}$	effective force
g	gravitational acceleration
h	particle distribution function
H	feature size
J	Jacobian
K_{eq}	equivalent heat conductivity
$\bar{K}_{eq}$	average equivalent heat conductivity
p	pressure
$\mathbf{r}$	position vector on physical plane
r_i, r_o	radius of annulus
t	time
T	temperature
T_0	reference temperature
ΔT	temperature difference

$\mathbf{u}$	velocity vector
U_1	mainstream velocity
x, y	coordinates on physical plane
ξ	position vector on computational plane
$\Delta \xi_{up,i}$	distance
ξ, η	coordinates on computational plane
χ	thermal diffusion coefficient
ν	kinematic coefficient of viscosity
β	expansion coefficient
ρ	density
Ω_i	collision term
τ	relaxation parameter
ω	weight coefficient
λ	CFL number
δt	time step

Subscripts

$\alpha, 1, 2$	coordinate direction indices
i	discrete number, inner surface of annulus
o	outer surface of annulus

Superscripts

eq	equilibrium
$*$	post-collision state

into the standard LBM by using a coarse grid in the whole domain and a finer grid in critical region [9,10]. Meanwhile, great efforts have been made in recent years to propose LBM schemes that can be implemented on a non-uniform computational grid. Among these schemes, which will be specifically addressed in this paper is the interpolation-supplemented lattice Boltzmann method (ISLBM) proposed by He & Doolen [11]. In the ISLBM, the flow domain is discretized into an arbitrary grid while a regular velocity lattice sits at each grid node. The ISLBM consists of three steps: collision, advection, and interpolation. The particle collision step is carried out at the grid node which is identical with LBM. However, because of the irregular computational grid distribution, the position of particle is generally no longer at a grid node after advection. Therefore, the interpolation step is needed to determine the value of particle distribution functions at each grid node. In order to keep the same accuracy as the LBM, the second-order upwind interpolation scheme or other high accurate schemes should be used.

Compared with LBM, the grid number of ISLBM has been reduced significantly and the computational efficiency is improved. Nevertheless, the ISLBM scheme still has its limitation, because it requires an analytical function for describing the transformations between physical plane and computational plane. Then, Imamura [12,13] carried out an in-depth research regarding the application of a body-fitted grid in LBM and developed the generalized form of interpolation-supplemented LBM (GILBM). The GILBM breaks through the limitation of ISLBM and the computational efficiency is improved as well [14]. It has been successfully used to simulate the flow around a circular cylinder and NACA0012 airfoil with good agreement with other results based on uniform grids. However, all these applications proceed under an isothermal condition. Once the velocity and temperature fields must be coupled with each other, the applicability of GILBM turns to be unknown. The purpose of this paper is to present a thermal LBM model in a body-fitted

coordinates by combining the thermal LBM model on uniform grids proposed by Inamuro et al. [15] with the GILBM scheme, and validates its effectiveness and availability with some benchmark solutions. Besides, different from the numerical investigation of mixed flow and heat transfer problems in irregular geometry by M. Sheikholeslami et al. [16,17], only single component fluid flow and heat transfer in body-fitted coordinates is focused, currently. The rest of this paper is organized as follows. In section 2, the thermal LBM model in a body-fitted coordinates is presented in detail. In section 3, the model has been applied to investigate the flow around a circular cylinder and natural convection in a horizontal cylindrical annulus. Characteristics of fluid flow and heat transfer, as well as the comparisons with previous computational and experimental data, are obtained. Finally, some conclusions are summarized in section 4.

2. Methodology**2.1. The generalized form of interpolation-supplemented LBM**

The LBM on orthogonal coordinate system describes fluid flows by the collision and advection calculation of particle distribution functions. In the present study, the two-dimensional nine-velocity (D2Q9) model is used. The governing equation of LBM, lattice Boltzmann equation (LBE), is described as

$$f_i(\mathbf{r} + \mathbf{e}_i \delta t, t + \delta t) - f_i(\mathbf{r}, t) = -\frac{1}{\tau_f} [f_i(\mathbf{r}, t) - f_i^{eq}(\mathbf{r}, t)] \quad (1)$$

where $i = 0, 1, 2, \dots, 8$. $f_i(\mathbf{r}, t)$ and $f_i^{eq}(\mathbf{r}, t)$ are particle distribution function and equilibrium distribution function corresponding to the discrete velocity vector $\mathbf{e}_i$ at position $\mathbf{r}$ and time t , τ_f is the relaxation time of LBE, δt is time step. The discrete velocities $\mathbf{e}_i$ are given as

$$(\mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4, \mathbf{e}_5, \mathbf{e}_6, \mathbf{e}_7, \mathbf{e}_8) = \begin{bmatrix} 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \end{bmatrix} \quad (2)$$

The equilibrium distribution functions can be constructed as

$$f_i^{eq} = \rho \omega_i \left[1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{2c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right] \quad (3)$$

where c_s is lattice sound speed, ω_i are weight coefficients whose values are given as

$$(\omega_0, \omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7, \omega_8) = \left(\frac{4}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{36}, \frac{1}{36}, \frac{1}{36}, \frac{1}{36} \right) \quad (4)$$

The macroscopic quantities such as density, velocity and pressure can be evaluated by

$$\rho = \sum_i f_i \quad (5)$$

$$\mathbf{u} = \frac{1}{\rho} \sum_i f_i \mathbf{e}_i \quad (6)$$

$$p = \rho c_s^2 \quad (7)$$

In order to develop a LBM model in a body-fitted coordinates, grid generation should be completed primarily. Generally, it can be done by solving a set of partial differential equations which has been detailed in Refs. [18,19]. The right and left sides of Eq. (1) correspond to the collision and advection terms respectively. In the implementation of the LBM, Eq. (1) can be divided into two processes as

$$f_i^*(\mathbf{r}, t) = f_i(\mathbf{r}, t) - \frac{1}{\tau_f} [f_i(\mathbf{r}, t) - f_i^{eq}(\mathbf{r}, t)] : \text{Collision term} \quad (8)$$

$$f_i(\mathbf{r}, t + \delta t) = f_i^*(\mathbf{r} - \mathbf{e}_i \delta t, t) : \text{Advection term} \quad (9)$$

where f_i^* is post-collision particle distribution function.

In the GILBM, there are two planes, physical plane and computational plane. In this paper, the physical plane and computational plane are denoted by $\mathbf{r} = (x_1, x_2) = (x, y)$ and $\xi = (\xi_1, \xi_2) = (\xi, \eta)$, respectively. Eqs. (8) and (9) on the physical plane should be transformed into the ones on the computational plane. As the collision process only involves the current grid node, the collision equations can be easily transformed into the computational plane by replacing $\mathbf{r}$ in Eq. (8) with ξ , thus, the equation can be rewritten as

$$f_i^*(\xi, t) = f_i(\xi, t) - \frac{1}{\tau_f} [f_i(\xi, t) - f_i^{eq}(\xi, t)] \quad (10)$$

After the collision process, the values of particle distribution functions at each grid node are acquired. While in the advection step, its governing equations are not so easy to be transformed. The reason is that the discrete velocities are variables on the computational plane which are different from the constant ones on the physical plane [12]. As shown in Fig. 1, the discrete velocity $\mathbf{e}_1 = (1, 0)$ keeps its value and direction unchanged on the physical plane but it varies with the nodes on the computational plane.

Hence, the contravariant velocity $\tilde{\mathbf{e}}_i = (\tilde{e}_{i,1}, \tilde{e}_{i,2})$ has been defined on computational plane, and its components are given as

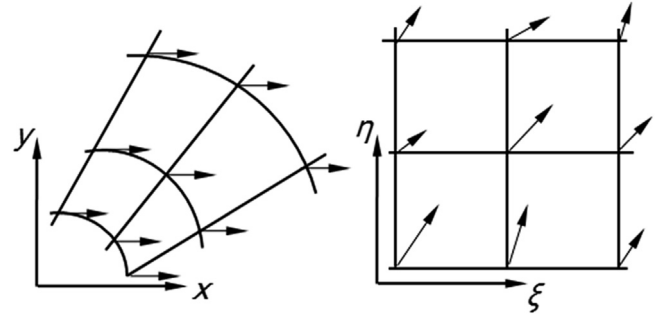


Fig. 1. Velocity vectors in different coordinates.

$$\tilde{e}_{i,1} = e_{i,1} \frac{\partial \xi_1}{\partial x_1} + e_{i,2} \frac{\partial \xi_1}{\partial x_2} \quad (11a)$$

$$\tilde{e}_{i,2} = e_{i,1} \frac{\partial \xi_2}{\partial x_1} + e_{i,2} \frac{\partial \xi_2}{\partial x_2} \quad (11b)$$

where $e_{i,1}$ and $e_{i,2}$ are components of the discrete velocity $\mathbf{e}_i$. In order to solve Eq. (11a) and Eq. (11b), a Jacobi conversion is needed as follows

$$\begin{bmatrix} \frac{\partial \xi_1}{\partial x_1} & \frac{\partial \xi_1}{\partial x_2} \\ \frac{\partial \xi_2}{\partial x_1} & \frac{\partial \xi_2}{\partial x_2} \end{bmatrix} = \frac{1}{J} \begin{bmatrix} \frac{\partial x_2}{\partial \xi_2} & -\frac{\partial x_1}{\partial \xi_2} \\ -\frac{\partial x_2}{\partial \xi_1} & \frac{\partial x_1}{\partial \xi_1} \end{bmatrix} \quad (12)$$

where the Jacobian J is calculated as

$$J = \frac{\partial x_1}{\partial \xi_1} \frac{\partial x_2}{\partial \xi_2} - \frac{\partial x_1}{\partial \xi_2} \frac{\partial x_2}{\partial \xi_1} \quad (13)$$

The differentials in Eqs. (12) and (13) are calculated through central-difference scheme with second-order accuracy. Once the contravariant velocity is obtained, the position where the distribution function comes from can be determined by integrating the contravariant velocity over δt . Then, the advection functions can be constructed as

$$f_i(\xi, t + \delta t) = f_i^*(\xi - \Delta \xi_{up,i}, t) \quad (14)$$

where $\Delta \xi_{up,i}$ is the advection distance, which can be given as

$$\Delta \xi_{up,i} = \int_0^{\delta t} d\xi_i = \int_0^{\delta t} \tilde{\mathbf{e}}_i dt \quad (15)$$

In order to get an accurate result, the calculation of advection distance is achieved by two-step Runge–Kutta method given as follows

$$\Delta \xi_{up,i}^{(1)} = \frac{1}{2} \delta t \tilde{\mathbf{e}}_i(\xi) \quad (16)$$

$$\Delta \xi_{up,i} = \delta t \tilde{\mathbf{e}}_i(\xi - \Delta \xi_{up,i}^{(1)}) + O(\delta t^3) \quad (17)$$

In Eqs. (14) and (17), the values of terms $f_i^*(\xi - \Delta \xi_{up,i}, t)$ and $\tilde{\mathbf{e}}_i(\xi - \Delta \xi_{up,i}^{(1)})$ cannot be directly gained as their positions are not located at the grid nodes on computational plane, so a interpolation procedure is extremely needed.

As for the internal lattice points, the second-order upwind interpolation method is chosen to ensure the numerical precision, and the interpolation function is given as

$$f_i(\xi, t + \delta t) = f_i^*(\xi - \Delta\xi_{up,i}, t) \\ = \sum_{m=0}^2 \sum_{l=0}^2 a_{i,m,2} a_{i,l,1} f_i^*(\xi_{i+m \cdot md}, \eta_{j+l \cdot nd}, t + \delta t) \quad (18)$$

where $md = \text{sgn}(\Delta\xi_{up,i,x})$ and $nd = \text{sgn}(\Delta\xi_{up,i,y})$ are coefficients determining the grid nodes for interpolation. sgn is a sign function. $a_{i,m,2}$ and $a_{i,l,1}$ are interpolation coefficients which can be calculated as

$$a_{i,0,\alpha} = \frac{1}{2}(|\Delta\xi_{up,i,\alpha}| - 1)(|\Delta\xi_{up,i,\alpha}| - 2) \\ a_{i,1,\alpha} = -|\Delta\xi_{up,i,\alpha}|(|\Delta\xi_{up,i,\alpha}| - 2) \\ a_{i,2,\alpha} = \frac{1}{2}|\Delta\xi_{up,i,\alpha}|(|\Delta\xi_{up,i,\alpha}| - 1) \quad (19)$$

For the nodes that adjacent the boundaries, central difference interpolation method is used because of the restriction of nodes number. The interpolation coefficients corresponding to the central difference interpolation method are calculated as

$$a_{i,0,\alpha} = \frac{1}{2}\Delta\xi_{up,i,\alpha}(\Delta\xi_{up,i,\alpha} - 1) \\ a_{i,1,\alpha} = 1 - \Delta\xi_{up,i,\alpha}^2 \\ a_{i,2,\alpha} = \frac{1}{2}\Delta\xi_{up,i,\alpha}(\Delta\xi_{up,i,\alpha} + 1) \quad (20)$$

In GILBM, the time step δt should be recalculated to guarantee the numerical stability. In this paper, it is determined by the Courant–Friedrichs–Lewy (CFL) condition, which is given as

$$\delta t = \lambda \min_{i,\alpha,\xi} \left| \frac{1}{\tilde{c}_{i,\alpha}|\xi|} \right| \quad (21)$$

where λ is CFL number ($0 < \lambda \leq 1$). It has been pointed out by Lee and Lin [20] whose essence is to guarantee that the particle should not exceed the area of surrounding four meshes after advection as shown in Fig. 2.

Although the calculation of contravariant velocity and its integration should be performed, it only need to be done once at the beginning of the simulation as these values do not change during the simulation. Furthermore, LBM and GILBM have been assessed by studying the fluid flow in irregular domains [14]. As the Reynolds number increases, the accuracy of simulation based on LBM reduces, while that of GILBM rises. Also, the time that GILBM taking to have the results from simulation is only 60% of that of LBM at low Reynolds number, and the proportion reduces as the Reynolds number. Given the higher computational efficiency of GILBM, the thermal LBM model in body-fitted coordinates has been proposed and validated in next sections.

2.2. The thermal LBM model in body-fitted coordinates

In order to simulate a non-isothermal flow problem, a thermal LBM model is needed. Compared with other thermal models in orthogonal coordinates, a simplest thermal governing equation has been introduced in Ref. [21]. In this section, a thermal LBM model in a body-fitted coordinates will be developed. Similar to the construction process of GILBM model, the thermal LBM model in

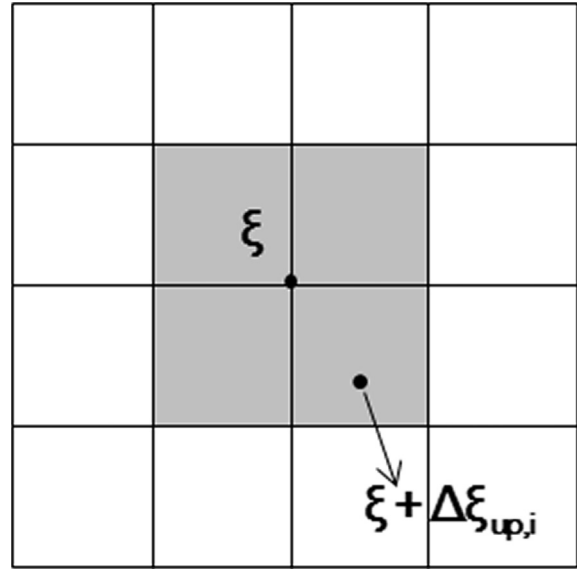


Fig. 2. The region of particle after advection on the computational plane.

orthogonal coordinates should also be divided into two processes, given as

$$h_i^*(\mathbf{r}, t) = h_i(\mathbf{r}, t) - \frac{1}{\tau_T} [h_i(\mathbf{r}, t) - h_i^{eq}(\mathbf{r}, t)] : \text{Collision term} \quad (22)$$

$$h_i(\mathbf{r}, t + \delta t) = h_i^*(\mathbf{r} - \mathbf{e}_i, t) : \text{Advection term} \quad (23)$$

where h_i is the thermal particle distribution function. h_i^{eq} is the thermal equilibrium distribution function, which can be calculated by

$$h_i^{eq} = \omega_i T \left[1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right] \quad (24)$$

h_i^* is the post-collision thermal distribution function, τ_T is the dimensionless relaxation time. Then the transformation of the collision and advection equations from physical plane to computational plane can be easily achieved as

$$h_i^*(\xi, t) = h_i(\xi, t) - \frac{1}{\tau_T} [h_i(\xi, t) - h_i^{eq}(\xi, t)] \quad (25)$$

$$h_i(\xi, t + \delta t) = h_i^*(\xi - \Delta\xi_{up,i}, t) \\ = \sum_{m=0}^2 \sum_{l=0}^2 a_{i,m,2} a_{i,l,1} h_i^*(\xi_{i+m \cdot md}, \eta_{j+l \cdot nd}, t + \delta t) \quad (26)$$

where coefficients and advection distance in Eq. (26) are all the same with former. The macroscopic temperature T can be calculated as

$$T = \sum_i h_i \quad (27)$$

In non-isothermal flow problems, the variance of temperature results in the change of velocity field, in other words, the temperature and velocity fields are coupled with each other. In order to

describe such situation, a force term F_i has been added to the right hand of Eq. (10) [22] which is given as

$$f_i^*(\xi, t) = f_i(\xi, t) - \frac{1}{\tau_f} [f_i(\xi, t) - f_i^{eq}(\xi, t)] + \delta t F_i \quad (28)$$

The form of force term F_i is variable under different conditions, but they all share a common essence, that is building the connection between the temperature and velocity fields. According to the Refs. [23,24], F_i is given as

$$F_i = \left(1 - \frac{1}{2\tau_f}\right) \omega_i \left[\frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})}{c_s^2} \mathbf{e}_i \right] \cdot \mathbf{G} \quad (29)$$

where $\mathbf{G}$ is the effective force, it can be calculated as

$$\mathbf{G} = -\beta(T - T_0)\mathbf{g} \quad (30)$$

where β is expansion coefficient, T_0 is reference temperature, $\mathbf{g}$ is the acceleration caused by external force.

It has been proved in the references [22,25] that through Chapman–Enskog expansion, the corresponding macroscopic equations in Cartesian coordinate can be recovered, and the kinematic coefficient of viscosity ν and thermal diffusion coefficient χ are determined, respectively, by

$$\nu = \frac{1}{3}c^2\left(\tau_f - \frac{1}{2}\right)\delta t, \quad \chi = \frac{1}{3}c^2\left(\tau_T - \frac{1}{2}\right)\delta t \quad (31)$$

where c is the lattice speed. Moreover, in the proposed model, the transformation of governing equations from the physical plane to the computational plane do not affect the calculation of transport coefficient. Therefore, the corresponding Prandtl number (Pr) can be tuned by the dimensionless relaxation time τ_f and τ_T as $Pr = \nu/\chi = (2\tau_f - 1)/(2\tau_T - 1)$.

A procedure of numerical calculation is summarized as follows:

- (1) Grid generation: Orthogonal grids with equal spacing are generated on the computational plane. The corresponding relation of grid points between two planes is determined by solving a set of partial differential equations of Elliptic Equation.
- (2) Initialization: Based on the physical problems, the values of density, velocity and temperature at each grid node are firstly initialized. Then, the initial values of distribution function $f_i(\mathbf{r}, 0)$ and $h_i(\mathbf{r}, 0)$ can be obtained.
- (3) Calculation of contravariant velocity $\mathbf{e}_i$, global time step δt and interpolation coefficients: Firstly, contravariant velocity $\mathbf{e}_i$ is determined by Eqs. (11)–(13) at each grid node. The interpolation coefficients can be calculated by Eqs. (19) and (20). Then, the global time step δt is obtained based on Eq. (21).
- (4) Collision: The collision process is carried out based on Eqs. (25) and (28) to determine the post-collision distribution functions $h_i^*(\xi, t)$ and $f_i^*(\xi, t)$.
- (5) Advection and interpolation: During the advection process, particle should move to a new position which is not located at the grid node, so the values of post-advection distribution functions $f_i(\xi, t + \delta t)$ and $h_i(\xi, t + \delta t)$ should be interpolated by Eqs. (18) and (26).
- (6) Calculation of macroscopic quantities: After the step (4–5), values of distribution functions $f_i(\xi, t)$ and $h_i(\xi, t)$ have been updated at each grid node. Then the density, velocity and pressure can be updated through Eqs. (5)–(7), and temperature is through Eq. (27).

- (7) Circulation: Go to step (4) until the desired computational precision is satisfied.

3. Numerical results and discussion

In this section, the proposed thermal LBM in body-fitted coordinates is validated by three cases including flow around a circular cylinder, natural convection in horizontal concentric and eccentric cylindrical annulus, respectively.

3.1. Flow around a circular cylinder

When fluid flows around a circular cylinder, the flow field would varies with Reynolds number (Re), which is defined as

$$Re = \frac{U_1 D}{\nu} \quad (32)$$

where U_1 is mainstream velocity, D is the diameter of circular cylinder. Fig. 3 shows the physical model of circular cylinder with $D = 20$, and the mainstream velocity $U_1 = 0.1$. The initial temperature of the cylindrical surface is set to be $T_0 = 1.0$ while the fluid is $T_1 = 0.0$. Three cases are performed in different simulated conditions in which Re is set as 10, 20 and 40, respectively. But, the Prandtl number (Pr) is set as 0.71 in all the three cases according to the Ref. [26]. Fig. 4 shows the relation between physical and computational planes. Non-equilibrium extrapolation boundary treatment are applied to the cylindrical surface and to the outer boundary, respectively. A periodic boundary condition is applied at the grid-cut in the wake.

First, a grid resolution study has been done for $Re = 20$. The wake length L and separation angle θ in different grid sizes have been listed in Table 1. It can be found that the results will not change significantly when the resolution has increased to 181×241 . So, the physical domain is divided by an O-grid with resolution of 181×241 for Reynolds number of 10 and 20. As for the Reynolds number of 40, similarly, the resolution is chosen as 226×301 . The dimensionless relaxation times τ_f and τ_T are determined by Eq. (31) and two important criterion numbers Re and Pr , as follow

$$\tau_f = \frac{3U_1 D}{Re \delta t} + 0.5, \quad \tau_T = \frac{3U_1 D}{Re Pr \delta t} + 0.5 \quad (33)$$

In general, a vortex tends to generate behind the circular cylinder. As Re increases, the vortex appears to be obvious, and its wake length and separation angle also increase [27]. The streamlines under different Re are shown in Fig. 5.

The pressure on the cylindrical surface has been calculated in its dimensionless form, which is expressed by the pressure coefficient C_p [11] with the form of

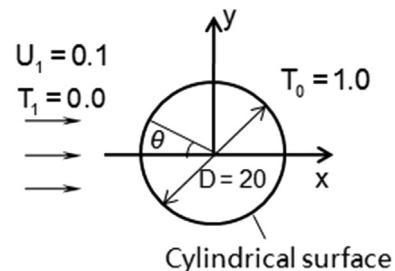


Fig. 3. Physical model of present problem.

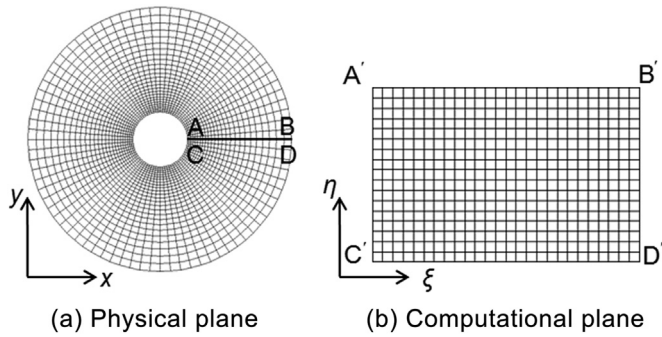


Fig. 4. The relation between (a) physical plane and (b) computational plane.

Table 1

The grid resolution study when $Re = 20$.

Physical quantity	61×121	121×181	181×241	241×301
$2L/D$	1.607	1.786	1.852	1.864
θ	41.94	42.56	42.89	42.97

$$C_p = \frac{p - p_\infty}{\frac{1}{2} \rho U^2} \quad (34)$$

where p is the pressure around the cylindrical surface, p_∞ is the pressure of far field. As shown in Fig. 6, the results are in good agreement with the ones in Ref. [11] which proves the applicability of the proposed model in the simulation of complex flow with large curvature boundaries.

The convective heat transfer coefficient on the cylindrical surface is described by local Nusselt number Nu and average Nusselt number $\bar{Nu}$ [26], which is calculated by

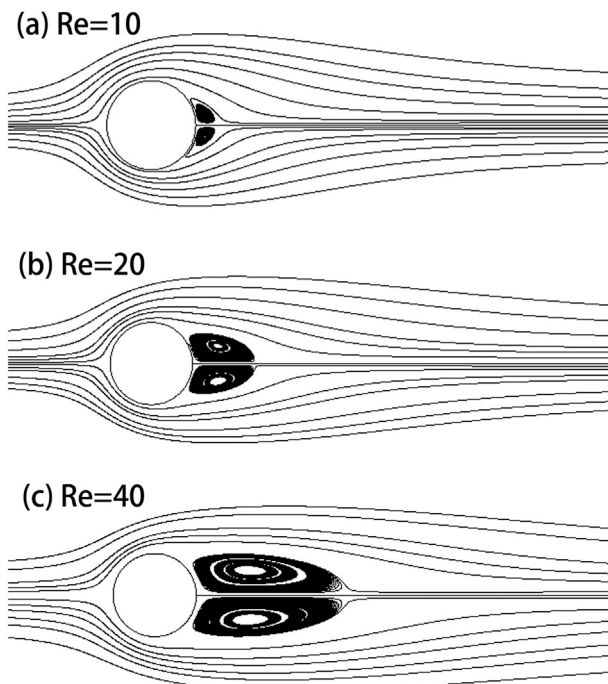


Fig. 5. Streamlines under different Reynolds numbers.

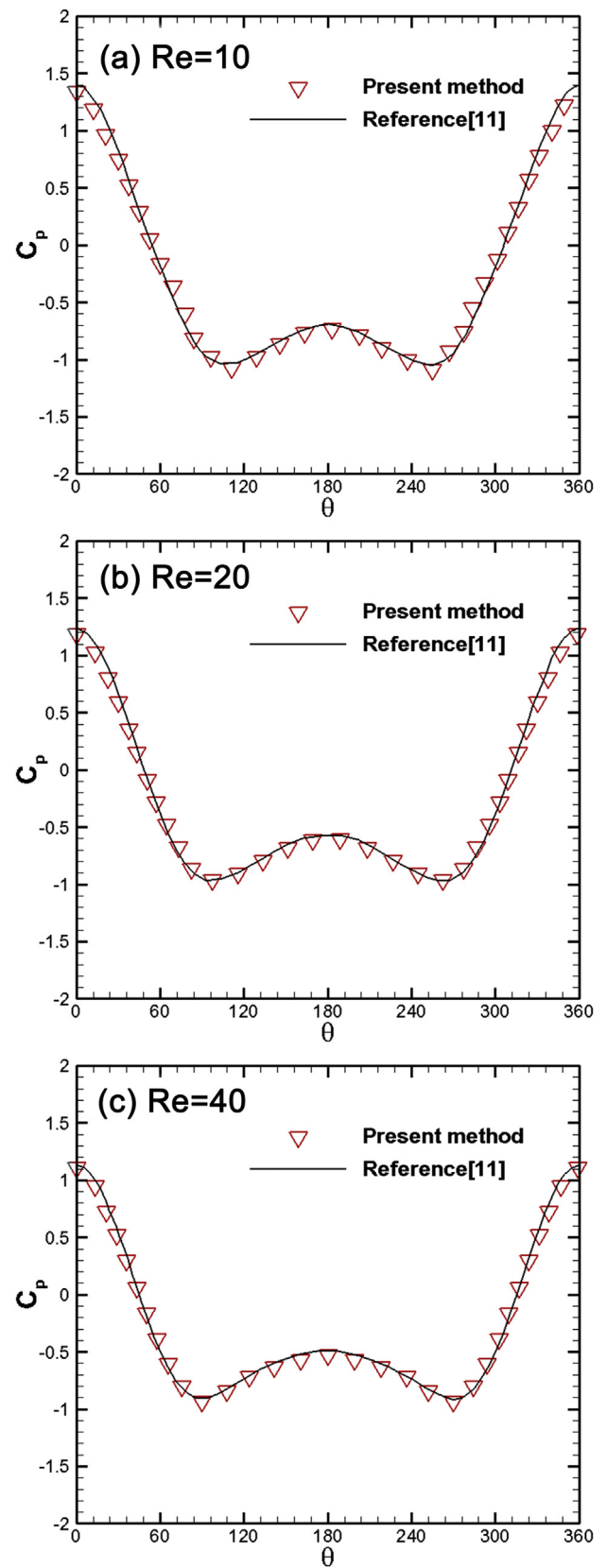


Fig. 6. Comparison of pressure distribution between present method and Ref. [11].

$$Nu = \frac{D}{(T_w - T_f)} \cdot \left(\frac{\partial T}{\partial n} \right)_{wall} \quad (35)$$

$$\overline{Nu} = \frac{1}{2\pi} \int_{\pi}^{-\pi} Nu d\theta \quad (36)$$

where T_w is the wall temperature, T_f is the temperature of far field.

$\left(\frac{\partial T}{\partial n} \right)_{wall}$ is temperature gradient on cylindrical surface.

Fig. 7 shows the temperature distribution around the circular cylinder when $Re = 40$. Local Nusselt number distribution around the cylindrical surface when $Re = 40$ is shown in Fig. 8. From Figs. 7 and 8, the characteristics of convective heat transfer can be found. As the front of circular cylinder directly scoured by the coming flow, the thermal boundary layer is thinnest at $\theta = 0^\circ$ which results in the biggest Nu there. However, because the thermal boundary layer gradually thickens along the flow direction, the Nu decreases accordingly. Such a tendency changes at the separation point because of the disturbance in the separation zone, and the Nu rises again. $\overline{Nu}$ of cylindrical surface under different Reynolds numbers, as well as comparison with previous computational and experimental results [28,29], are listed in Table 2. It can be seen that all the three results agree well.

3.2. Natural convection in horizontal concentric cylindrical annulus

Natural convective heat transfer from a body to a finite space enclosing it has received considerable attention for a long time [30–32]. Fig. 9 shows the physical model of a horizontal concentric cylindrical annulus with $r_i = 10$ and $r_o/r_i = 2.6$. The initial temperatures of outer and inner surface are set to be $T_o = 0.0$ and $T_i = 1.0$, respectively. Firstly, the Rayleigh numbers (Ra), which influence the natural convective heat transfer in the annulus, is defined as

$$Ra = \frac{g\beta\Delta TH^3 Pr}{\nu^2} \quad (37)$$

where ΔT is the temperature difference between high and low temperature wall, H is the feature size. Because the computational domain is symmetrical, half of it has been analyzed and the relation between physical and computational planes is shown in Fig. 10. A body-fitted grid (C-grid) with grid resolution of 81×201 is used in the present simulations. Non-equilibrium extrapolation boundary treatment is implemented on all the boundaries. Three cases are carried out in which the Rayleigh numbers are set as 10^3 , 10^4 , 5×10^4 , respectively. The Prandtl number of the air enclosed in the annular space is set to be fixed as 0.7 which is corresponding to the condition of previous experiment [33]. Here, the thermal diffusion coefficient is chosen to be 0.04, then, the dimensionless relaxation times (τ_f and τ_T) are easy to obtain.

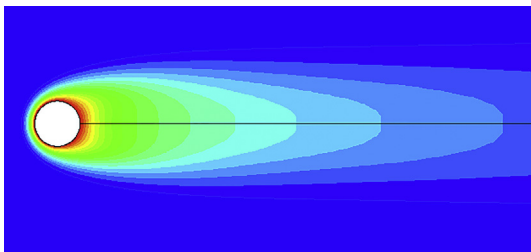


Fig. 7. Temperature distribution around the circular cylinder when $Re = 40$.

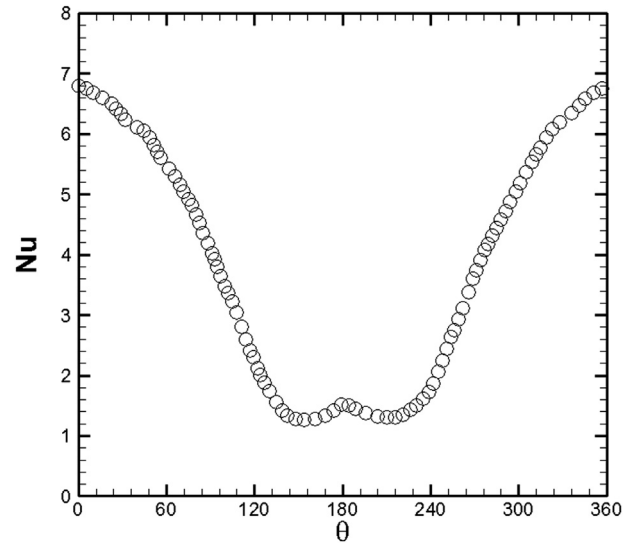


Fig. 8. Local Nusselt number distributions around the cylindrical surface when $Re = 40$.

Table 2

Comparison of $\overline{Nu}$ around cylindrical surface under different Reynolds numbers.

Method	$\overline{Nu}$		
	$Re = 10$	$Re = 20$	$Re = 40$
Cengel Y A [28]	1.97	2.57	3.36
Churchill and Bernstein [29]	1.85	2.48	3.38
Present	2.08	2.65	3.47

The heat transfer effect between the annulus has been measured by equivalent heat conductivity K_{eq} [33] which is defined as

$$K_{eq} = (\text{Convective heat})/(\text{Conductive heat}) \quad (38)$$

Hence, the local equivalent heat conductivity of inner and outer surfaces can be calculated as

$$K_{eqi} = \frac{r_i \cdot \ln(r_o/r_i)}{T_i - T_o} \cdot \left. \frac{\partial T}{\partial r} \right|_{r=r_i} \quad (39)$$

$$K_{eqo} = \frac{r_o \cdot \ln(r_o/r_i)}{T_i - T_o} \cdot \left. \frac{\partial T}{\partial r} \right|_{r=r_o} \quad (40)$$

As for average equivalent heat conductivity $\overline{K}_{eqi}$ and $\overline{K}_{eqo}$, the only difference is temperature gradient in Eqs. (39) and (40) which should be replaced by

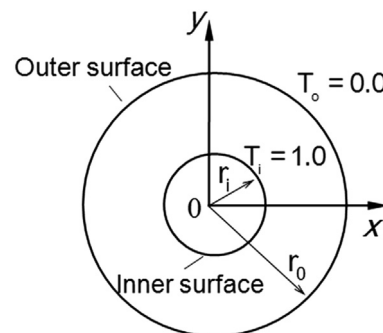


Fig. 9. Physical model of horizontal concentric cylindrical annulus.

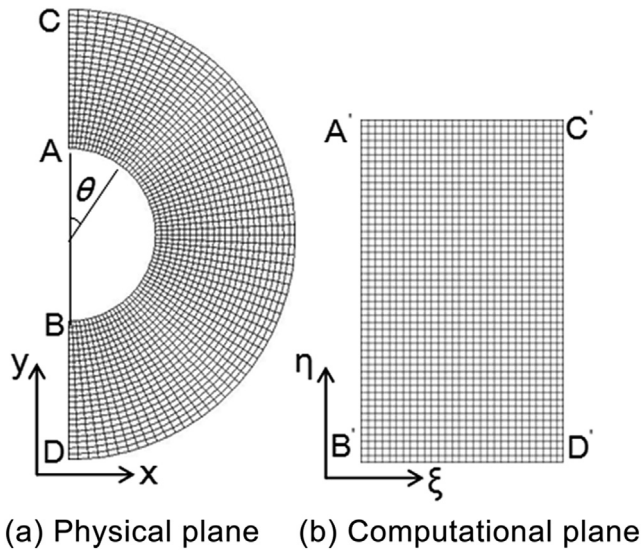


Fig. 10. The relation between (a) physical plane and (b) computational plane.

$$\frac{\partial \bar{T}}{\partial r} = \frac{1}{\pi} \int_0^\pi \frac{\partial T}{\partial r} \cdot \partial \theta \quad (41)$$

where $\frac{\partial \bar{T}}{\partial r}$ is the average temperature gradient around cylindrical surface.

Fig. 11 shows the streamlines between concentric annulus under different Rayleigh numbers. The fluid enclosed in the annulus generates a vortex. When the Rayleigh number is small, the fluid flows weakly, and the center of the vortex is almost located at middle of the annulus. With the increasing of Rayleigh number, the center of the vortex gradually moves up, and the flowing at the top of the annular space changes to be strong while turns to be weak at the bottom. Fig. 12 shows the comparison of temperature distribution between the experiment [34] and present method when $Ra = 5 \times 10^4$. Their isotherm present S shape and the distributions are similar to each other which can demonstrate the rationality of the proposed model further.

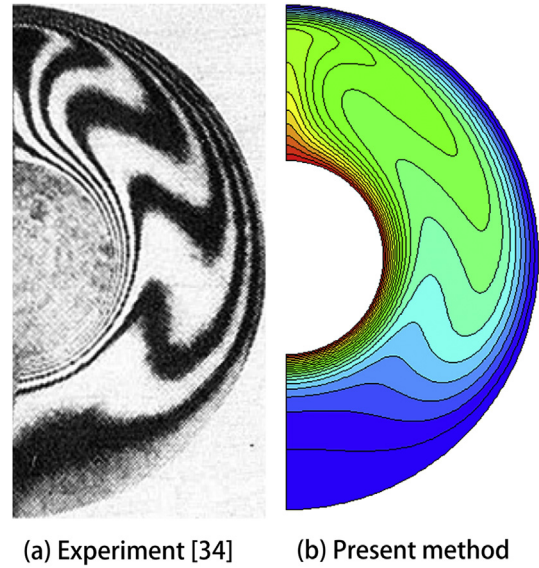


Fig. 12. Comparison of temperature distribution between (a) experiment and (b) present method in concentric annulus when $Ra = 5 \times 10^4$.

Furthermore, the local equivalent heat conductivity calculated by present method when $Ra = 5 \times 10^4$, as well as comparisons with previous experimental data in Ref. [33], are shown in Fig. 13. It can be seen that they are in a good agreement. The largest difference between the experimental and numerical values is less than 10.2% and the average values are within 3.3% of each other. Table 3 lists the average equivalent heat conductivity data which agree well again. The relative errors tend to increase with Ra while the maximum value is less than 5% and about 3% for average value.

3.3. Natural convection in horizontal eccentric cylindrical annulus

The flow and heat transfer in the eccentric horizontal annulus have also been simulated in this section. Fig. 14 shows the physical model of horizontal eccentric annulus. Boundary conditions are all set the same with concentric annulus in previous section except the location of inner cylinder. ϵ is the offset distance of inner cylinder

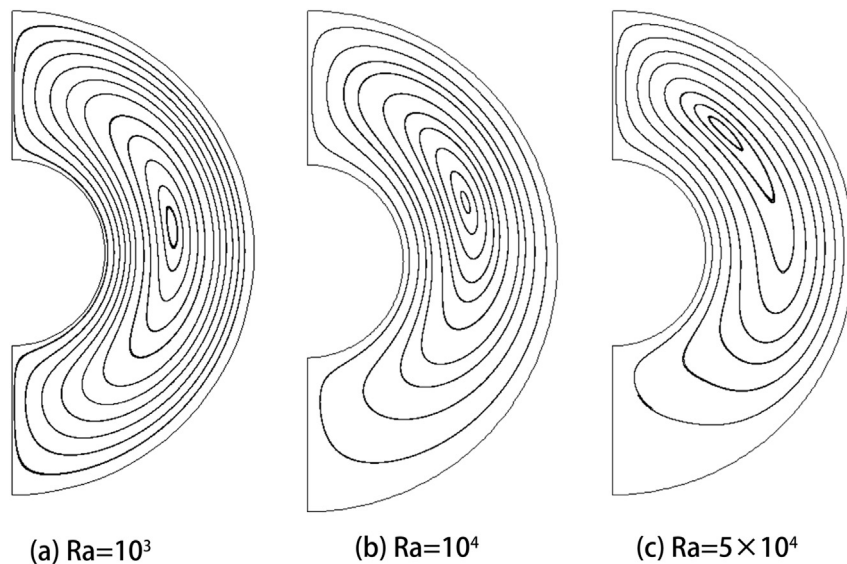


Fig. 11. Streamlines between concentric annulus under different Rayleigh numbers.

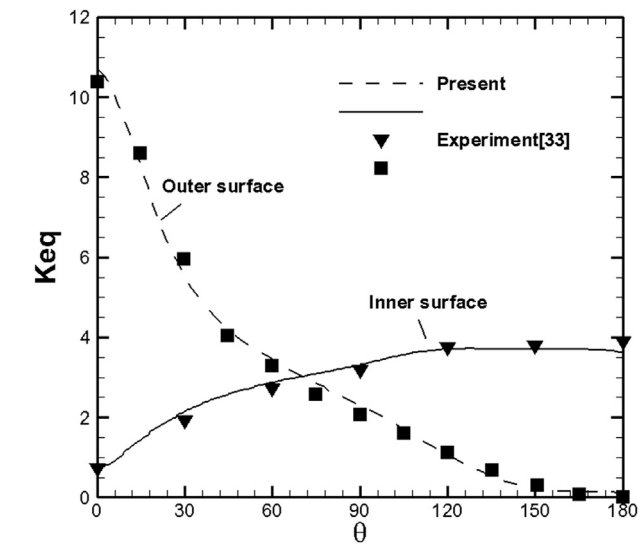


Fig. 13. Comparison of experimental and numerical local equivalent conductivity in concentric annulus when $Ra = 5 \times 10^4$.

Table 3
Comparison of average equivalent conductivity for different Rayleigh numbers.

Ra	$\bar{K}_{eq}$					
	Inner surface			Outer surface		
	Ref. [33]	Present	Error	Ref. [33]	Present	Error
10^3	1.081	1.060	1.9%	1.084	1.062	2.0%
10^4	2.010	2.062	2.6%	2.005	2.063	2.8%
5×10^4	3.036	3.148	3.7%	3.088	3.152	2.1%

which is set as $\epsilon/L = 0.623$, $L = r_o - r_i$. Such a symmetrical domain is divided into 81×201 grid nodes as well, and Fig. 15 shows relation between physical and computational planes.

When $Ra = 5 \times 10^4$, the streamlines in the eccentric annulus are shown in Fig. 16. Apparently, as the upper annular space is wider than the lower part, the location of vortex center is upper than that of concentric annulus under the same condition. Fig. 17 shows the comparison of temperature distribution between experiment [34] and present method. It can be found that present result is also well matched with the experimental one.

The comparison of experimental [34] and numerical results of local equivalent conductivity in eccentric annulus when $Ra = 5 \times 10^4$ has been shown in Fig. 18. Their variations are consistent, and the largest error between the two results is about

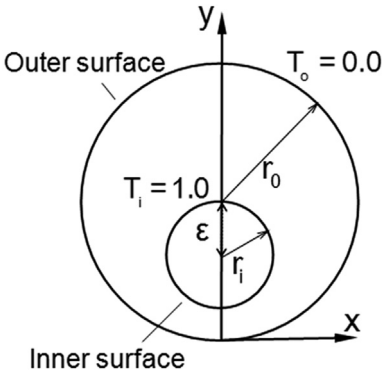


Fig. 14. Physical model of horizontal eccentric cylindrical annulus.

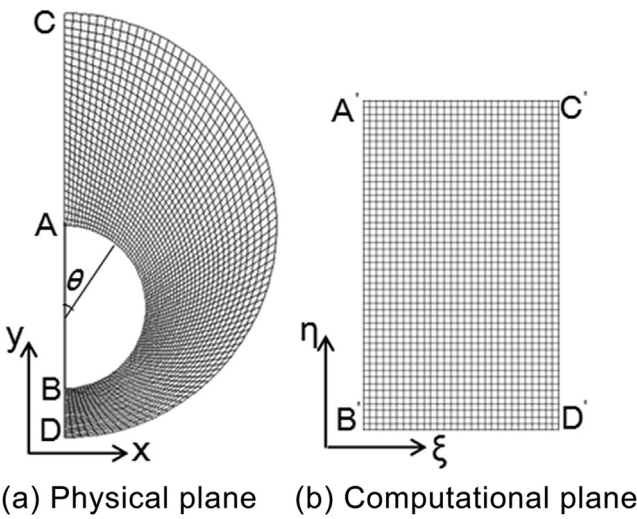


Fig. 15. The relation between (a) physical plane and (b) computational plane.

7.12% while the average one is less than 3%. The good agreement results demonstrate once again that the proposed model is reliable.

4. Conclusion remarks

In this paper, a thermal LBM model in body-fitted coordinates has been proposed by combining the GILBM scheme and the thermal LBM model on uniform grid of Inamuro et al. [12–15]. Specifically, the construction of thermal distribution functions and the coupling of velocity and temperature fields in body-fitted coordinates are stated. Firstly, the proposed model has been used to investigate flow around a circular cylinder under different Reynolds numbers. In addition to velocity and temperature distributions, the pressure coefficient and Nusselt number around cylindrical surface

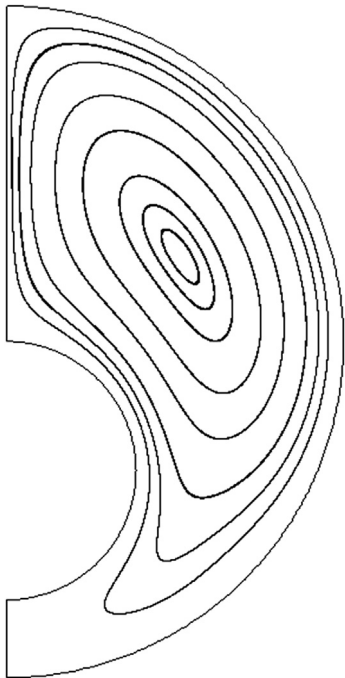


Fig. 16. Streamlines in eccentric annulus when $Ra = 5 \times 10^4$.

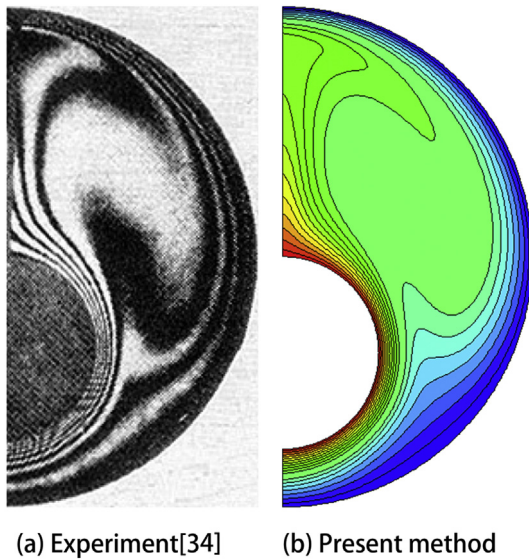


Fig. 17. Comparison of temperature distribution between (a) experiment and (b) present method in eccentric annulus when $Ra = 5 \times 10^4$.

have been obtained which get a good agreement with previous studies. It is found that both pressure coefficient and Nusselt number increase with Reynolds number. Secondly, the proposed model is applied to simulate natural convection in horizontal concentric cylindrical annulus. Three cases with various Rayleigh numbers are studied. Heat convection strengthens gradually as Rayleigh number increases which can be demonstrated by the rising of equivalent heat conductivity. The variation of equivalent heat conductivity agrees well with the result in Ref. [33]. Finally, when $Ra = 5 \times 10^4$, the heat transfer between eccentric annuli has been calculated which once again agrees well with experimental result [34]. All in all, the above studies indicate that it is feasible and reliable to apply the present model to study complex flow and heat transfer problems with large curvature boundaries in a body-fitted coordinates. However, in the present paper, only single component fluid flow and heat transfer in body-fitted

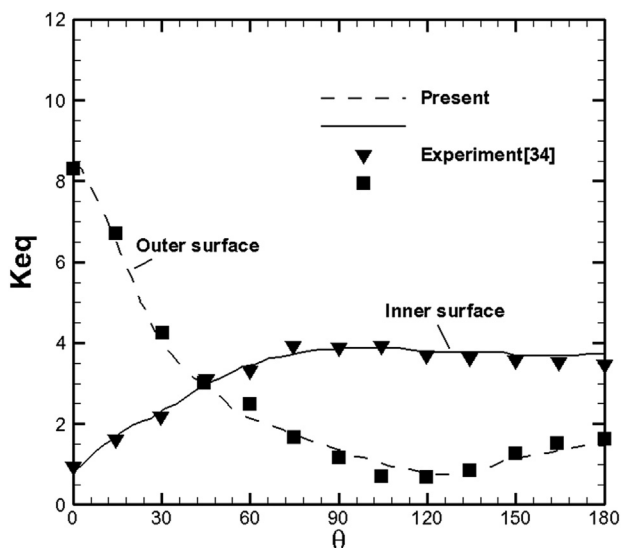


Fig. 18. Comparison of experimental and numerical local equivalent conductivity in eccentric annulus when $Ra = 5 \times 10^4$.

coordinates is focused, and neither turbulence nor flow mixed is considered, which deserves further investigation in the future.

Acknowledgments

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